

# Package ‘twopiece’

May 5, 2026

**Type** Package  
**Title** The family of twopiece distributions  
**Version** 0.1.0  
**Author** F. Javier Rubio  
**Maintainer** The package maintainer <fxrubio@gmail.com>  
**Description** Density, distribution function, quantile function and random generation for the 3-parameter twopiece distribution with 3 parameterizations: two-piece (tp), epsilon-skew (eps), and inverse scale factors (isf).  
**License** GPL-3  
**Encoding** UTF-8  
**LazyData** true  
**RoxygenNote** 7.3.3

## Contents

|                |          |
|----------------|----------|
| dtp3 . . . . . | 1        |
| dtp4 . . . . . | 2        |
| ptp3 . . . . . | 3        |
| ptp4 . . . . . | 3        |
| qtp3 . . . . . | 4        |
| qtp4 . . . . . | 4        |
| rtp3 . . . . . | 5        |
| rtp4 . . . . . | 6        |
| <b>Index</b>   | <b>7</b> |

---

|      |                                                                            |
|------|----------------------------------------------------------------------------|
| dtp3 | <i>Probability Density Function for 3-parameter twopiece distributions</i> |
|------|----------------------------------------------------------------------------|

---

## Description

Probability Density Function for 3-parameter twopiece distributions

## Usage

dtp3(x, mu, par1, par2, FUN, param = "tp", log = FALSE)

**Arguments**

|        |                                                                                          |
|--------|------------------------------------------------------------------------------------------|
| log    | log.p: logical; if TRUE, probabilities p are given as log(p).                            |
| x:     | vector of quantiles.                                                                     |
| p:     | vector of probabilities.                                                                 |
| n:     | number of observations. If length(n) > 1, the length is taken to be the number required. |
| mu:    | location parameter.                                                                      |
| par1:  | scale parameter 1.                                                                       |
| par2:  | scale parameter 2.                                                                       |
| FUN:   | a symmetric density f.                                                                   |
| param: | parameterizations used.                                                                  |

dtp4

*Probability Density Function for 4-parameter twopiece distributions***Description**

Probability Density Function for 4-parameter twopiece distributions

**Usage**

```
dtp4(x, mu, par1, par2, delta, FUN, param = "tp", log = FALSE)
```

**Arguments**

|        |                                                                                          |
|--------|------------------------------------------------------------------------------------------|
| log    | log.p: logical; if TRUE, probabilities p are given as log(p).                            |
| x:     | vector of quantiles.                                                                     |
| p:     | vector of probabilities.                                                                 |
| n:     | number of observations. If length(n) > 1, the length is taken to be the number required. |
| mu:    | location parameter.                                                                      |
| par1:  | scale parameter 1.                                                                       |
| par2:  | scale parameter 2.                                                                       |
| delta: | shape parameter.                                                                         |
| FUN:   | a symmetric density f.                                                                   |
| param: | parameterizations used.                                                                  |

---

|      |                                                                               |
|------|-------------------------------------------------------------------------------|
| ptp3 | <i>Cumulative Probability Function for 3-parameter twopiece distributions</i> |
|------|-------------------------------------------------------------------------------|

---

**Description**

Cumulative Probability Function for 3-parameter twopiece distributions

**Usage**

```
ptp3(x, mu, par1, par2, FUN, param = "tp", log.p = FALSE)
```

**Arguments**

|        |                                                                                          |
|--------|------------------------------------------------------------------------------------------|
| x:     | vector of quantiles.                                                                     |
| p:     | vector of probabilities.                                                                 |
| n:     | number of observations. If length(n) > 1, the length is taken to be the number required. |
| mu:    | location parameter.                                                                      |
| par1:  | scale parameter 1.                                                                       |
| par2:  | scale parameter 2.                                                                       |
| FUN:   | a symmetric density f.                                                                   |
| param: | parameterizations used.                                                                  |
| log    | log.p: logical; if TRUE, probabilities p are given as log(p).                            |

---

|      |                                                                               |
|------|-------------------------------------------------------------------------------|
| ptp4 | <i>Cumulative Probability Function for 4-parameter twopiece distributions</i> |
|------|-------------------------------------------------------------------------------|

---

**Description**

Cumulative Probability Function for 4-parameter twopiece distributions

**Usage**

```
ptp4(x, mu, par1, par2, delta, FUN, param = "tp", log.p = FALSE)
```

**Arguments**

|       |                                                                                          |
|-------|------------------------------------------------------------------------------------------|
| x:    | vector of quantiles.                                                                     |
| p:    | vector of probabilities.                                                                 |
| n:    | number of observations. If length(n) > 1, the length is taken to be the number required. |
| mu:   | location parameter.                                                                      |
| par1: | scale parameter 1.                                                                       |
| par2: | scale parameter 2.                                                                       |

delta: shape parameter.  
 FUN: a symmetric density f.  
 param: parameterizations used.  
 log: log.p: logical; if TRUE, probabilities p are given as log(p).

---

qtp3

*Quantile Function for 3-parameter twopiece distributions*


---

### Description

Quantile Function for 3-parameter twopiece distributions

### Usage

```
qtp3(p, mu, par1, par2, FUN, param = "tp")
```

### Arguments

x: vector of quantiles.  
 p: vector of probabilities.  
 n: number of observations. If length(n) > 1, the length is taken to be the number required.  
 mu: location parameter.  
 par1: scale parameter 1.  
 par2: scale parameter 2.  
 FUN: a symmetric density f.  
 param: parameterizations used.  
 log: log.p: logical; if TRUE, probabilities p are given as log(p).

---

qtp4

*Quantile Function for 4-parameter twopiece distributions*


---

### Description

Quantile Function for 4-parameter twopiece distributions

### Usage

```
qtp4(p, mu, par1, par2, delta, FUN, param = "tp")
```

**Arguments**

|        |                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------|
| x:     | vector of quantiles.                                                                               |
| p:     | vector of probabilities.                                                                           |
| n:     | number of observations. If $\text{length}(n) > 1$ , the length is taken to be the number required. |
| mu:    | location parameter.                                                                                |
| par1:  | scale parameter 1.                                                                                 |
| par2:  | scale parameter 2.                                                                                 |
| delta: | shape parameter.                                                                                   |
| FUN:   | a symmetric density f.                                                                             |
| param: | parameterizations used.                                                                            |
| log    | log.p: logical; if TRUE, probabilities p are given as $\log(p)$ .                                  |

rtp3

*Random Number Generation Function for 3-parameter twopiece distributions***Description**

Random Number Generation Function for 3-parameter twopiece distributions

**Usage**

```
rtp3(n, mu, par1, par2, FUN, param = "tp")
```

**Arguments**

|        |                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------|
| x:     | vector of quantiles.                                                                               |
| p:     | vector of probabilities.                                                                           |
| n:     | number of observations. If $\text{length}(n) > 1$ , the length is taken to be the number required. |
| mu:    | location parameter.                                                                                |
| par1:  | scale parameter 1.                                                                                 |
| par2:  | scale parameter 2.                                                                                 |
| FUN:   | a symmetric density f.                                                                             |
| param: | parameterizations used.                                                                            |
| log    | log.p: logical; if TRUE, probabilities p are given as $\log(p)$ .                                  |

---

|      |                                                                                 |
|------|---------------------------------------------------------------------------------|
| rtp4 | <i>Random Number Generation Function for 4-parameter twopiece distributions</i> |
|------|---------------------------------------------------------------------------------|

---

**Description**

Random Number Generation Function for 4-parameter twopiece distributions

**Usage**

```
rtp4(n, mu, par1, par2, delta, FUN, param = "tp")
```

**Arguments**

|        |                                                                                          |
|--------|------------------------------------------------------------------------------------------|
| x:     | vector of quantiles.                                                                     |
| p:     | vector of probabilities.                                                                 |
| n:     | number of observations. If length(n) > 1, the length is taken to be the number required. |
| mu:    | location parameter.                                                                      |
| par1:  | scale parameter 1.                                                                       |
| par2:  | scale parameter 2.                                                                       |
| delta: | shape parameter.                                                                         |
| FUN:   | a symmetric density f.                                                                   |
| param: | parameterizations used.                                                                  |
| log    | log.p: logical; if TRUE, probabilities p are given as log(p).                            |

# Index

dtp3, [1](#)  
dtp4, [2](#)

ptp3, [3](#)  
ptp4, [3](#)

qtp3, [4](#)  
qtp4, [4](#)

rtp3, [5](#)  
rtp4, [6](#)